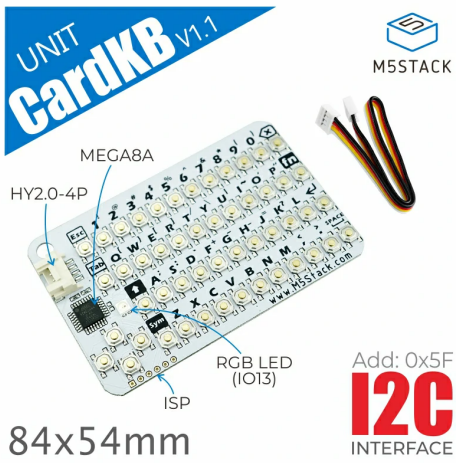
Unit CardKB v1.1

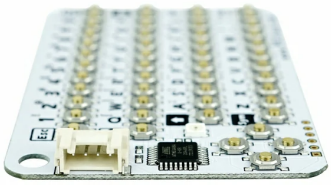
SKU:U035-B

















Description

Unit CardKB v1.1 is a card-sized QWERTY **50-key** PCB matrix keyboard that uses **ATMega8A** as the encoding MCU, with an output interface of **I2C**. It has one onboard **RGB-LED** to indicate keyboard status.

Features

- Full keyboard input with multiple key combinations.
- HY2.0-4P interface, supports UIFlow and Arduino.

Includes

- 1 x Unit CardKB v1.1
- 1 x HY2.0-4P Grove connection cable (20cm)

Applications

- Keyboard peripheral for M5Stack Core

Specifications

Specification	Parameter
MCU	ATMega8A
Number of Keys	50
RGB LED	1 pc
Communication Interface	I2C Communication @0x5F
Product Size	88.0 x 54.0 x 5.0mm
Product Weight	14.3g
Package Size	170.0 x 120.0 x 7.0mm
Gross Weight	18.0g

Learn

- **Press a single key**, the keyboard will output the first key value (letter keys will output in lowercase). For example, pressing "Q" will output "q" (lowercase).
- **Sym+key**, the keyboard will output the second key value. For example, after clicking "Sym", pressing "Q" will output "{". Double-click "Sym" to lock the function, and all subsequent key presses will output the second key value. Double-click "Sym" again to unlock.
- **Shift+key**, the keyboard will output the uppercase form of the letter. For example, after clicking "Shift", pressing "Q" will output "Q" (uppercase). Double-click "Shift" to lock the function, and all subsequent key presses will output in uppercase. Double-click "Shift" again to unlock.
- **Fn+key (custom function key combination)**, the keyboard will output the third key value. You can customize the function of the pressed key.

Line number	Button combination													
Line 1	Key	ESC	1	2	3	4	5	6	7	8	9	0	Back	Up
	Value	0x1B	0x31	0x32	0x33	0x34	0x35	0x36	0x37	0x38	0x39	0x30	0x08	0xB5
Line 2	Key	TAB	q	w	e	r	t	y	u	i	o	p	Fn	Down
	Value	0x09	0x71	0x77	0x65	0x72	0x74	0x79	0x75	0x69	0x6F	0x70	NULL	0xB6
Line 3	Key		Shift	a	s	d	f	g	h	j	k	l	Enter	Left
	Value		NULL	0x61	0x73	0x64	0x66	0x67	0x68	0x6A	0x6B	0x6C	0x0D	0xB4
Line 4	Key		Sym	z	x	c	v	b	n	m	,	.	SPACE	Right
	Value		NULL	0x7A	0x78	0x63	0x76	0x62	0x6E	0x6D	0x2C	0x2E	0x20	0xB7

Line number	Button combination													
Line 1	Sym+Key	ESC	!	@	#	$	%	^	&	*	(	)	Back	Up
	Value	0x1B	0x21	0x40	0x23	0x24	0x25	0x5E	0x26	0x2A	0x28	0x29	0x08	0xB5
Line 2	Sym+Key	TAB	{	}	[	]	/	\	|	–	'	"	Fn	Down
	Value	0x09	0x7B	0x7D	0x5B	0x5D	0x2F	0x5C	0x7C	0x7E	0x27	0x22	NULL	0xB6
Line 3	Sym+Key		Shift	;	:	`	+	-	_	=	?	NULL	Enter	Left
	Value		NULL	0x3B	0x3A	0x60	0x2B	0x2D	0x5F	0x3D	0x3F	NULL	0x0D	0xB4
Line 4	Sym+Key		Sym	NULL	NULL	NULL	NULL	NULL	NULL	NULL	<	>	SPACE	Right
	Value		NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	0x3C	0x3E	0x20	0xB7

Line number	Button combination													
Line 1	Shift+Key	ESC	1	2	3	4	5	6	7	8	9	0	Del	Up
	Value	0x1B	0x31	0x32	0x33	0x34	0x35	0x36	0x37	0x38	0x39	0x30	0x7F	0xB5
Line 2	Shift+Key	TAB	Q	W	E	R	T	Y	U	I	O	P	Fn	Down
	Value	0x09	0x51	0x57	0x45	0x52	0x54	0x59	0x55	0x49	0x4F	0x50	NULL	0xB6
Line 3	Shift+Key		Shift	A	S	D	F	G	H	J	K	L	Enter	Left
	Value		NULL	0x41	0x53	0x44	0x46	0x47	0x48	0x4A	0x4B	0x4C	0x0D	0xB4
Line 4	Shift+Key		Sym	Z	X	C	V	B	N	M	,	.	SPACE	Right
	Value		NULL	0x5A	0x58	0x43	0x56	0x42	0x4E	0x4D	0x2C	0x2E	0x20	0xB7

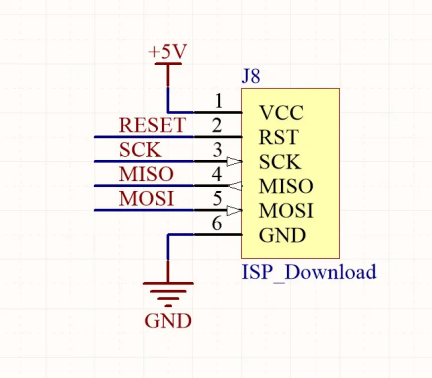
Line number	Button combination													
Line 1	Fn+Key	ESC	1	2	3	4	5	6	7	8	9	0	Back	Up
	Value	0x80	0x81	0x82	0x83	0x84	0x85	0x86	0x87	0x88	0x89	0x8A	0x8B	0x99
Line 2	Fn+Key	TAB	Q	W	E	R	T	Y	U	I	O	P	Fn	Down
	Value	0x8C	0x8D	0x8E	0x8F	0x90	0x91	0x92	0x93	0x94	0x95	0x96	NULL	0xA4
Line 3	Fn+Key		Shift	A	S	D	F	G	H	J	K	L	Enter	Left
	Value		NULL	0x9A	0x9B	0x9C	0x9D	0x9E	0x9F	0xA0	0xA1	0xA2	0xA3	0x98
Line 4	Fn+Key		Sym	Z	X	C	V	B	N	M	,	.	SPACE	Right
	Value		NULL	0xA6	0xA7	0xA8	0xA9	0xAA	0xAB	0xAC	0xAD	0xAE	0xAF	0xA5

PinMap

Unit CardKB v1.1

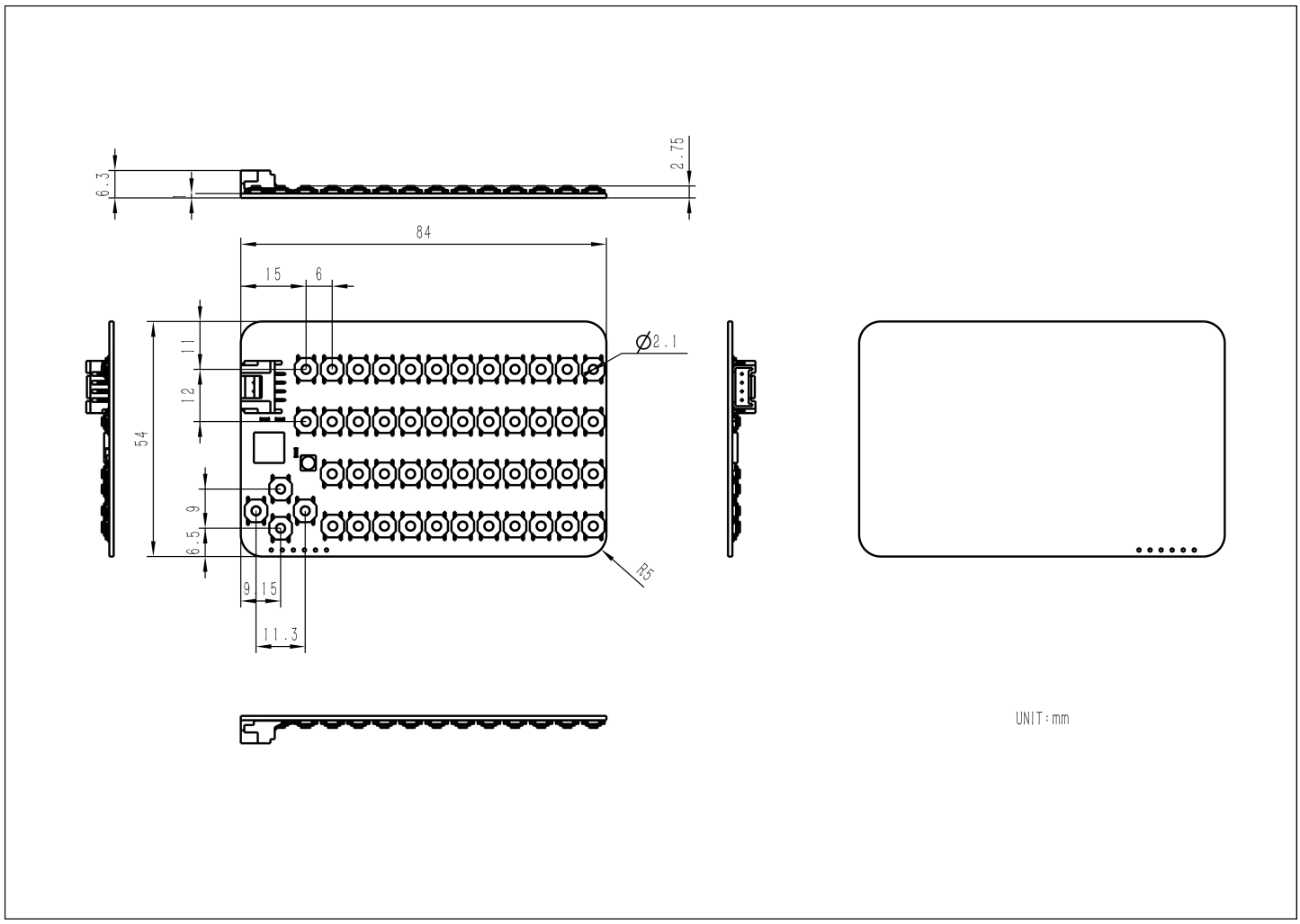
HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

ATMega8A ISP Download Interface Pin Definition



Model Size

Unit CardKB v1.1 Model Size PDF



Softwares

Arduino

- Unit CardKB v1.1 Example with M5Core



UiFlow1

- Unit CardKB v1.1 UiFlow1 Docs

UiFlow2

- Unit CardKB v1.1 UiFlow2 Docs

Internal Firmware

- Unit CardKB v1.1 Internal Firmware

Protocol

- Unit CardKB v1.1 Communication Protocol

EasyLoader

Easyloader	Download Link	Note
Unit CardKB example with M5Core	download	/

Video

M5stack Cardkb.mp4